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A new benchmark dataset for machine learning applications in discrete manufacturing: CiP-DMD

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Abstract

The development of machine learning applications in manufacturing depends primarily on the availability of meaningful and reliable data. In this paper, we present the Center for industrial Productivity - Discrete Manufacturing Dataset (CiP-DMD), the first open-source discrete manufacturing dataset of a multi-step machining process for which detailed documentation is available. The dataset encompasses a comprehensive description of the individual machining processes, the quality controls and the traceability concept used to match the process and quality data corresponding to the production of 847 pneumatic cylinders. With this new benchmark dataset, we aim to contribute to the development of new approaches and foment the permeation of machine learning in manufacturing.

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1. Introduction

Machine learning (ML) finds its use in production for performing various tasks, such as product quality prediction, condition monitoring and anomaly detection, adaptive production planning and control, and remaining useful lifetime prediction [1]. Manufacturing companies already using ML to assist these tasks report positive experiences with proofs of concept and actual application deployments that yield economic benefits [2]. However, the availability of suited datasets limits the development and testing of new applications since companies often see process and quality control data as sensitive information and therefore do not publicly share the training and testing datasets [3]. Benchmark datasets are essential for the evaluation of new approaches and for making the data and results available to the scientific community and thus advancing the practical implementation of ML in

manufacturing [4]. Motivated by the lack of suitable benchmark datasets encompassing multi-step production processes and the necessary traceability to link process and quality control data, we present the Center for industrial Productivity - Discrete Manufacturing Dataset (CiP-DMD). The dataset was recorded in the CiP learning factory, a production facility of the Institute for Production Management, Technology and Machine Tools (PTW) at the Technical University of Darmstadt in Germany. In the CiP, two components, i.e., piston rod and cylinder bottom are machined and assembled with supplier parts to manufacture a pneumatic cylinder. The CiP-DMD consists of data recordings of the manufacturing and quality control of 847 pneumatic cylinders. It comprises the numerical process data as well as metadata corresponding to the steps of the production process and traceability data, allowing for tracking of the components through the manufacturing process and its labels from the

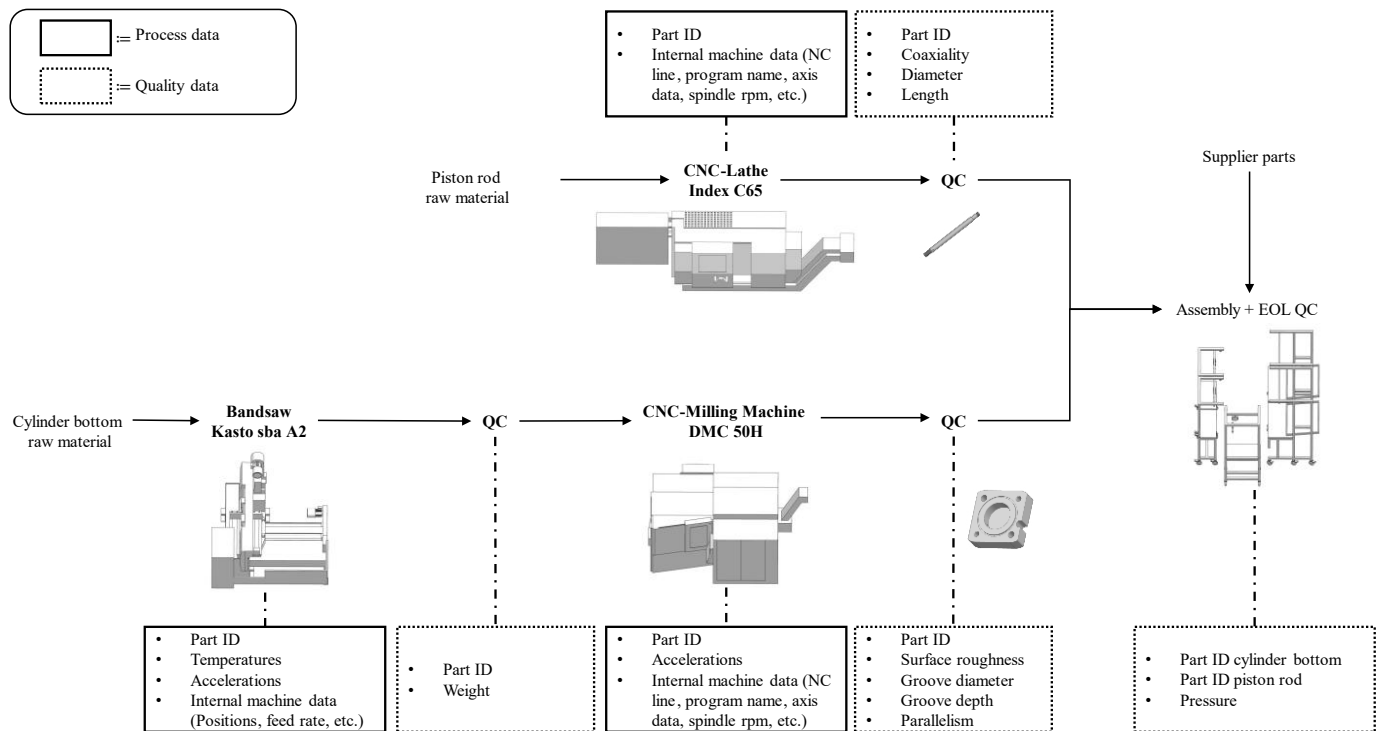


Figure 1: Overview of the individual manufacturing steps and the recorded data at each stage in the manufacturing chain of a pneumatic cylinder.

quality controls. The dataset is suited for investigating various supervised and unsupervised ML approaches for diversified manufacturing use cases, e.g., predictive quality and anomaly detection.

In this paper, we introduce the CiP-DMD and the details of how it was recorded and consolidated into a publicly available dataset. Section 2 provides the related work of existing open-source manufacturing datasets, highlighting the need for the presented dataset. Section 3 describes the manufacturing process and the details of the data acquisition and consolidation efforts. The paper concludes by pointing out future work directions and potential uses of the CiP-DMD dataset in Section 4.

2. Related work

Publicly accessible datasets capturing manufacturing processes are generally sparse when compared to other ML application fields, such as autonomous driving [3]. Companies often regard process data as sensitive information that cannot be shared due to privacy and competition-related concerns [5]. While sample quantity and quality are important for the performance of ML algorithms, existing datasets are often found lacking in both regards. This section lists existing datasets with use cases similar to CiP-DMD. We refer the interested reader to recent review publications such as [3,5] for a high-level comparison of existing datasets related to general ML in manufacturing.

In 2022, a CNC milling dataset was released by Bosch [6]. It features a multistage cutting process of aluminum parts and corresponding triaxial accelerometer data acquired with a

sampling rate of 2 kHz using a sensor mounted on the spindle housing. The dataset is partitioned into tool operations showing the production process and binary quality control labels of around 150 parts. While the Bosch CNC dataset is, to the best of the authors' knowledge, the only dataset that specifically addresses process monitoring/quality prediction, other datasets are available that capture CNC milling data for tool wear estimation purposes. Notable examples are the SMART Lab milling dataset [7], the PHM 2010 challenge dataset [8] and the NASA milling dataset [9]. As the datasets mentioned above were recorded in laboratory conditions and not during the manufacturing of actual products, no product quality labels are available.

In addition to CNC milling process data, the CiP-DMD dataset contains additional data related to turning and sawing, as well as machine control data unavailable in existing datasets. Detailed information regarding the manufactured part specifications and the actual Numerical Control (NC) programs are released with the process data for in-depth analysis. Furthermore, the CiP-DMD covers a multi-step manufacturing process, allowing for analysis of process interdependencies and their impact on the product quality.

3. CiP-DMD manufacturing process and data acquisition

In this section, we present the CiP-DMD along with the individual manufacturing steps and the data acquisition scheme involved in recording the dataset.

Figure 1 provides an overview of the manufacturing chain of a pneumatic cylinder and the recorded data, starting from the respective raw materials to the final assembly. The

manufacturing chain can be partitioned into three sub-chains: (1) piston rod, (2) cylinder bottom and (3) final assembly. We recorded data on the manufacturing and quality control steps along each sub-chain to obtain a holistic dataset.

The CiP-DMD comprises labelled data for 847 pneumatic cylinders recorded between November 2022 and June 2023. For 697 pneumatic cylinders, the available data represent normal process conditions, i.e., no anomalies are present within the manufacturing chain. Conversely, for 150 pneumatic cylinders, certain elements of the manufacturing chain are corrupted by anomalies. These anomalies primarily target the milling process and, to some extent, the sawing process in the manufacturing sub-chain of a cylinder bottom. We restricted the anomalies to these machines as they have been retrofitted with external sensors that provide a richer data source for ML applications such as, e.g. anomaly detection and predictive quality.

3.1. Traceability concept and dataset structure

In order to assign the process and quality data to their respective physical parts, we developed a traceability concept. This concept is based on quick response (QR) codes that are applied to a part and scanned at the respective stages in the manufacturing chain. The QR codes encode part IDs that we combine with the time stamps of the scans to establish anchor points. These anchor points allow us to map the data to the corresponding parts. The recorded process and quality data are stored separately for each sub-chain in a standardized folder structure. Figure 2 provides an excerpt of this structure for the piston rod sub-chain. Note that this structure is representative of both the cylinder bottom and final assembly data.

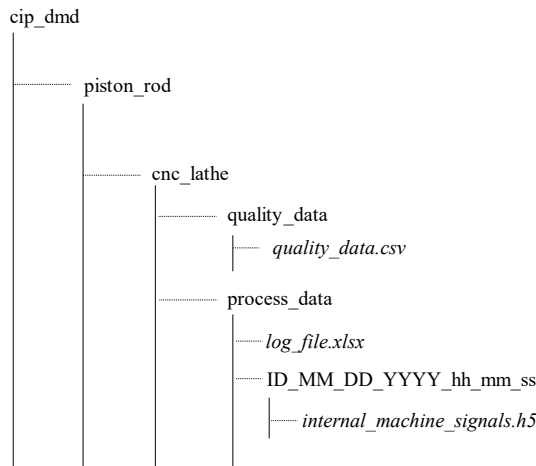


Figure 2: Excerpt of the folder structure and naming convention for the piston rod sub-chain used in the CiP-DMD.

The quality data is stored in a .csv file. This file contains the part IDs, the time stamps of the respective quality control, and the labels, i.e., “0” in case of a normal part and “1” in case of an anomalous part. The process data is stored in separate folders. These folders follow a naming convention in which a part ID is combined with the time stamp that marks the start of

the respective manufacturing step. We used 6-digit numbers as the part ID for the piston rods and cylinder bottoms.

Regarding the ID of a finished pneumatic cylinder, we concatenate the corresponding cylinder bottom and piston rod IDs used in the assembly. Thus, the part ID of a completed pneumatic cylinder is a 12-digit number. The respective process data is stored in Hierarchical Data Format (HDF)-5 file format. Note that we distinguish between internal and external machine signals depending on the type of sensors equipped in the respective machine. In addition to the process data, we included a .xlsx file containing logging information regarding replacing worn tools. Specifically, for each replacement of a worn tool, we provide the time stamp and the ID of the part that was first processed with the new tool.

In the following, we present further details regarding the individual manufacturing sub-chains and corresponding quality control steps.

3.2. Manufacturing sub-chain: Piston rod

The manufacturing sub-chain of the piston rods includes a CNC turning process and a quality control step.

3.2.1. CNC turning process

The CNC lathe used for producing the piston rods is an Index C65 machine. Figure 3 displays the test bed and gives an inside view of the machining space.

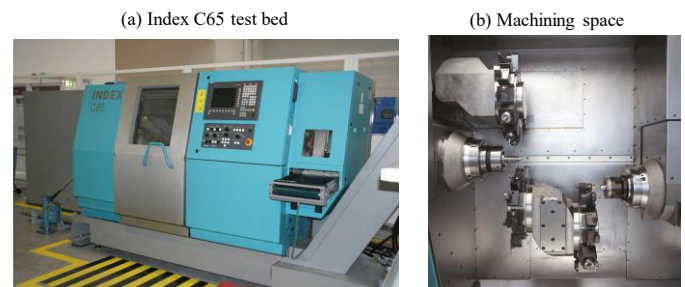


Figure 3: (a) Index C65 CNC lathe used to produce the piston rods. (b) Inside view of the machining space.

The piston rods consist of stainless steel that is delivered in 750 mm long entities to the machine. Within a single run of the CNC program, the machine processes the 750 mm rod and outputs four piston rods. The processing time of a machine cycle amounts to 436 s. In Figure 4, we present an image of a piston rod prior to and after processing. Since we obtain four piston rods from one machine cycle, we attach and scan the QR codes sequentially after processing. The Index C65 is not equipped with any external sensors. However, we use the Siemens Brownfield Connectivity (BFC) gateway to access internal machine control data via the Message Queuing Telemetry Transport (MQTT) protocol. Concretely, with the help of the BFC gateway, we gain access to data such as, e.g. the current NC line, spindle currents and axis data using a sampling rate of 5 Hz.



Figure 4: Piston rod of a pneumatic cylinder. (Left) Unprocessed raw material. (Right) Processed piston rod.

3.2.2. Quality control – CNC turning process

The processed piston rods are consequently passed to quality control. Here, we scan the QR code of the individual part and measure coaxiality, diameter and length. We rely on Hexagon/Q-DAS as the quality control software as it provides an interface for the Mahr measurement devices. Figure 5 shows the quality station for the piston rods.

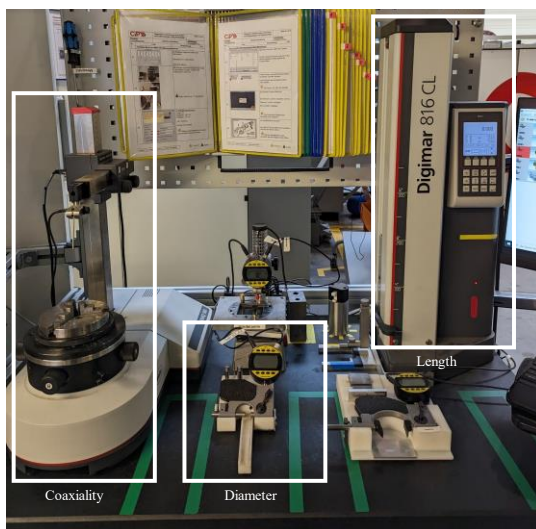


Figure 5: Quality control of the CNC turning process. (Left) Coaxiality meter. (Center): Dial gauge for the diameter. (Right): Altimeter to measure the length.

3.3. Manufacturing sub-chain: Cylinder bottom

The production of a cylinder bottom encompasses four steps. First, a piece of raw material is cut from a square bar of steel. Second, the sawn part is assigned a QR code sticker and passed to quality control to measure its weight. Third, the front and back side of the sawn part are processed in a CNC milling process to finish the cylinder bottom. Fourth, the completed cylinder bottom is passed to the second quality control, where the final quality characteristics, i.e., surface roughness, groove diameter, groove depth, and parallelism, are measured.

3.3.1. Sawing process

As a test bed for the sawing process, we use a retrofitted Kasto sba A2 machine. The Kasto sba A2 is equipped with various external sensors that record temperatures, accelerations, and other machine data with a sampling rate of 5 Hz. This data is continuously sent to a database via OPC Unified Architecture (OPC UA). The machine's processing time amounts to roughly 143s for a single part. As stated before, we corrupted the sawing process with controlled anomalies. Specifically, we cut a total of 99 parts too short. This anomaly corresponds to one of the two process anomalies used in the subsequent milling process. Cutting off a part too short results in a lower weight of the sawn part.

3.3.2. Quality control – sawing process

Once the sawing process is completed, the sawn part is passed to the first quality control. Here, we attach and scan the QR code while measuring the weight of the sawn part using an electronic scale. We access the scale measurement via OPC UA and store it with the label, part ID and time stamp in the respective .csv file.

3.3.3. CNC milling process

The next step involves the CNC milling process. For this purpose, we use a Deckel Maho DMC-50H horizontal CNC milling machine; see Figure 6 for an illustration of the test bed.



Figure 6: DMC-50H horizontal CNC milling machine used as the test bed for processing a cylinder bottom.

The machining space of the DMC-50H contains a rotating tower equipped with special fixtures to hold multiple parts at once. We use a single clamping on each tower side and process two distinct parts per machining cycle. Specifically, the rotating tower is equipped with a piece of raw material on the front side and a semi-finished part on the back side. Upon completion of the machining cycle, the machine outputs a semi-finished and a finished part. Figure 7 visualizes the rotating tower and the respective clamping positions. The processing time of a machining cycle is approximately 5 minutes. More precisely, it takes 199 s to complete the front side and 113 s to finish the back side of a cylinder bottom. See Figure 8 for a display of the corresponding production stages.

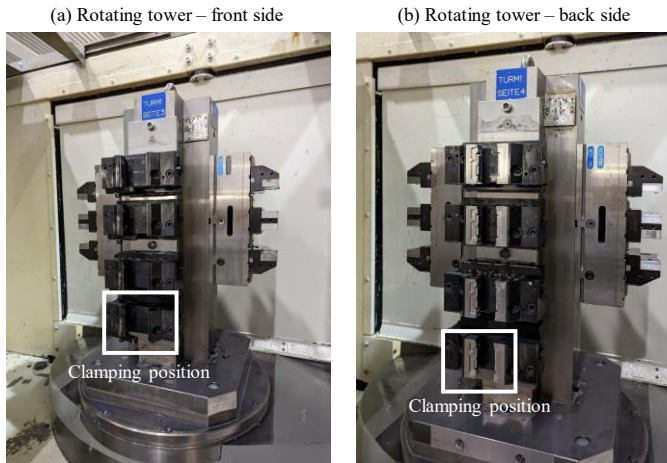


Figure 7: Rotating tower of the DMC-50H test bed. (a) Front side: Clamping of raw material. (b) Back side: Clamping of a semi-finished part.

The CNC milling process represents the most extensive part of the CiP-DMD. We provide labelled data for both normal and anomalous process conditions of internal and external machine signals. Regarding external machine signals, the DMC-50H has been retrofitted with a triaxial accelerometer mounted on the spindle and connected to an industrial PC on top of the machine. The accelerometer has a sampling rate of 2.5 kHz. In terms of the internal machine signals, such as e.g., NC line and spindle current, we use the BFC gateway as we did in the CNC turning process.

We corrupted the CNC milling process with two anomalies. The first anomaly targets the front side of the cylinder bottom and uses the 99 parts cut off too short in the sawing process. Processing a part that is too short causes the face milling step not to remove any chips, resulting in a significant decrease in the measured accelerations. Conversely, the second anomaly aims at the back side of a cylinder bottom and corresponds to a false clamping of another 40 parts. This anomaly causes the face milling step to remove more chips from one side than another, leading to a crooked cylinder bottom. The anomalies were designed in such a way that they affect specific quality characteristics in the subsequent quality control. Figure 9 provides a comparison of the respective normal and anomalous parts.

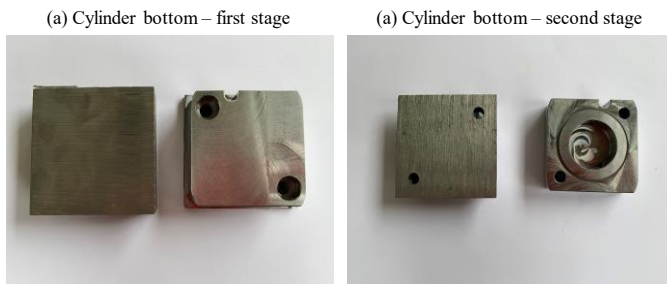


Figure 8: Cylinder bottom production stages. (a) First stage: conversion of raw material to semi-finished part. (b) Second stage: Conversion of the semi-finished part to the finished part.

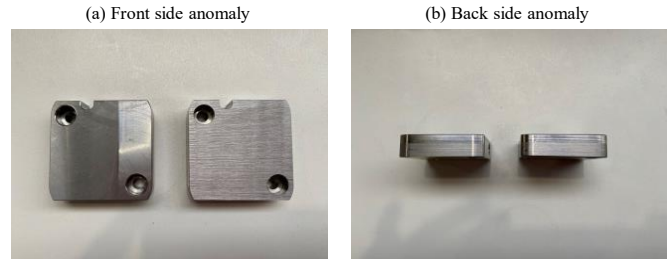


Figure 9: Anomalies used in the CNC milling process of a cylinder bottom. (a) Front side anomaly: Normal part (left) vs. anomalous part (right). (b) Back side anomaly: Normal part (left) vs. anomalous part (right).

Aside from these 139 controlled anomalies, a total of 11 miscellaneous anomalies occurred that were unplanned and correspond to, e.g., wrong process settings. Note that the anomalies in this process collectively account for the 150 anomalous parts in the CiP-DMD.

3.3.4. Quality control – CNC milling process

The last step in the cylinder bottom manufacturing sub-chain corresponds to quality control. In this process, we scan the QR code of the individual part and measure surface roughness, groove diameter, groove depth and parallelism. Similar to the quality control of the piston rods, we rely on Mahr measurement devices and the Hexagon/Q-DAS software to compare the measurements with the tolerance fields of the predefined quality characteristics. As stated before, the anomalies in the CNC milling process were designed to affect specific quality characteristics. Concretely, the first anomaly type affects surface roughness, while the second affects parallelism and groove depth. In Figure 10, we show the quality station for the cylinder bottoms.

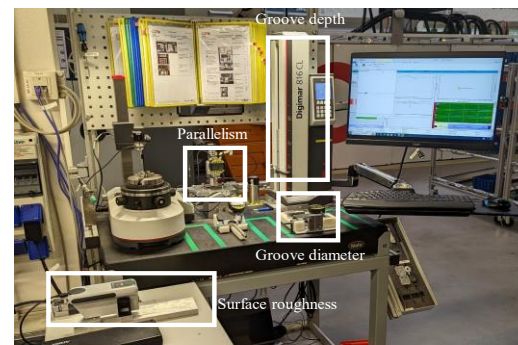


Figure 10: Quality control of the CNC milling process. (Left): Surface roughness device. (Center): Dial gauge for parallelism. (Top-right): Altimeter to measure groove depth. (Bottom-right): Dial gauge to measure groove diameter.

3.4. Manufacturing sub-chain: Final assembly

With the piston rods and cylinder bottoms completed, the last step in the manufacturing chain encompasses the final assembly and the end-of-line quality control (EOL QC). First, we use supplier parts, such as, e.g. casing, piston and sealing

rings, to assemble the pneumatic cylinder. Recall that we combine the part IDs of the cylinder bottom and piston rod to obtain the ID of the pneumatic cylinder. Once the pneumatic cylinder is assembled, we conduct a pressure test using a load cell that sends data via OPC UA. It is important to emphasize that the anomalies in the previous process steps do not necessarily affect the final pressure test. However, since 99 pneumatic cylinders are unprocessed from one side while another 40 are crooked and 11 contain various other errors, we nonetheless have 150 scrap cylinders. Figure 11 illustrates the individual components of a pneumatic cylinder and the setting for the final pressure test.

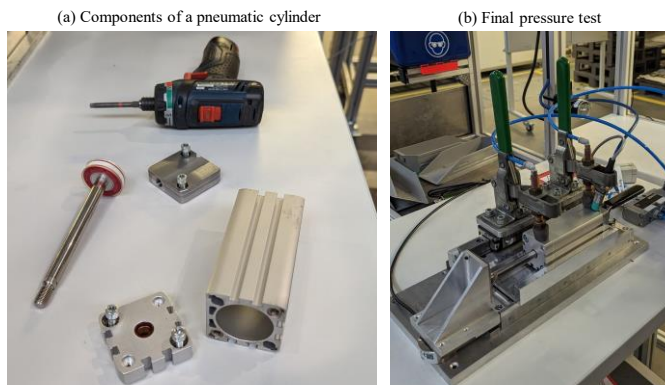


Figure 11: (a) Cylinder bottom, piston rod and supplier parts used to assemble a pneumatic cylinder. (b) Set up for the final pressure test.

4. Conclusion and outlook

In this paper, we introduced the CiP-DMD dataset, a comprehensive and openly available discrete manufacturing dataset comprising the process-, quality- and metadata from a multi-step manufacturing process of 847 pneumatic cylinders. We identified three major shortcomings of current discrete manufacturing datasets that we aim to address: (i) Existing discrete manufacturing datasets do not release internal machine control information corresponding to external sensors' measurements. This prohibits the linkage of, e.g., accelerometer measurements to machine states and NC commands, hindering further analysis. We address this limitation by releasing the NC programs, internal machine data and synchronized external sensor data for in-depth analysis and use in ML-related data processing. (ii) Existing discrete manufacturing datasets do not cover multi-step production processes that span across multiple machines, prohibiting the analysis of interdependencies between the processes. We address this limitation by releasing the data of three production machines as well as the final assembly, comprising the manufacturing chain of the pneumatic cylinder. (iii) While datasets exist that provide quality control labels for

manufacturing parts or geometries, these are usually binary OK/NOK values. To allow for a fine-grained analysis, we release the actual quality feature measurements as floating-point values, along with the corresponding control limits.

The dataset is available for research purposes online at the following link: <http://cipdmd.ptw-darmstadt.de>. The dataset may be used to investigate multiple discrete manufacturing-related ML tasks in the future, including quality prediction, anomaly detection as well as tool condition monitoring.

Acknowledgements

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